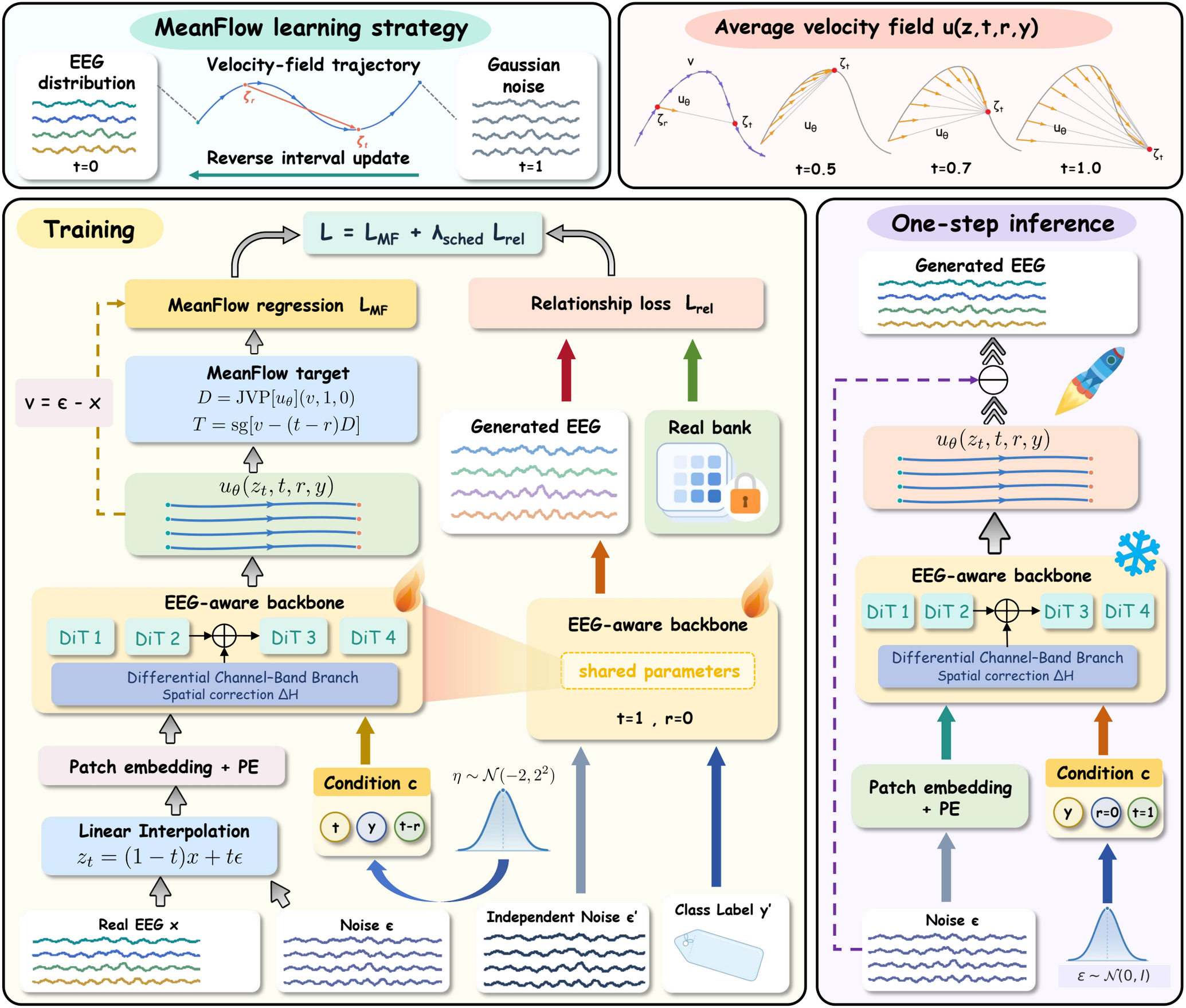

MeanFlow learning strategy
EEG distribution
t=0
Velocity-field trajectory
Reverse interval update
Gaussian noise
t=1
Average velocity field u(z,t,r,y)
t=0.5
t=0.7
t=1.0
Training
L = L_MF + λ_sched L_rel
MeanFlow regression L_MF
Relationship loss L_rel
MeanFlow target
D = JVP[u_θ](v, 1, 0)
T = sg[v − (t − r)D]
v = ϵ - x
u_θ(z_t, t, r, y)
Generated EEG
Real bank
EEG-aware backbone
DiT 1
DiT 2
DiT 3
DiT 4
Differential Channel-Band Branch
Spatial correction ΔH
shared parameters
t=1 , r=0
Patch embedding + PE
Linear Interpolation
z_t = (1 − t)x + tϵ
Condition c
t
y
t-r
η ∼ N(−2, 2²)
Real EEG x
Noise ϵ
Independent Noise ϵ'
Class Label y'
One-step inference
Generated EEG
u_θ(z_t, t, r, y)
EEG-aware backbone
DiT 1
DiT 2
DiT 3
DiT 4
Differential Channel-Band Branch
Spatial correction ΔH
Patch embedding + PE
Condition c
y
r=0
t=1
Noise ϵ
ε ∼ N(0, I)